MIL-C-21067B
28 JUNE 1966
SUPERSEDING
MIL-C-21067A(SHIPS)
23 OCTOBER 1964
(SEE 6.4)

MILITARY SPECIFICATION

COATING COMPOUND, SYNTHETIC RUBBER, FOR EXPOSED STEEL SURFACES

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers synthetic rubber coating compound intended for use in protecting exposed steel surfaces from corrosion.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-96 — Cans, Metal, 28 Gage and Lighter.

PPP-P-704 — Pails, Shipping, Steel (1 Through 12 Gallon).

MILITARY

MIL-P-699 — Paint, Deck, Gray, Outside.

MIL-E-15129 — Enamel, Ship, Exterior, Alkyd, Navy Gray, No. 7 (Formula No. 5N).

STANDARDS

FEDERAL

FED-STD-601 — Rubber: Sampling and Testing.

MILITARY

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

FSC 8030

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INTERSTATE COMMERCE COMMISSION

CFR Title 49, Parts 71–78.

(Copies may be obtained from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Copies may be obtained from the Official Classification Committee, 1 Par Avenue at 33rd Street, New York, New York 10016)

3. REQUIREMENTS

3.1 Preproduction sample. Prior to beginning production a sample shall be tested as specified in 4.2 (see 6.3).

3.2 Material. The coating material shall be a polymerized chloroprene compound dispersed in a solvent suitable for application by brush to either horizontal or vertical steel surfaces. A primer may be used as part of the coating system to obtain the required adhesion (see 3.3).

3.3 Physical requirements. The coating compound shall conform to the requirements specified in table I, when tested as specified in 4.6.

TABLE I. *Physical requirements for coating compound.*

	Requirement	Test procedure
Viscosity, variation from preproduction test value, percent, maximum	25	4.6.1
Solids content, percent, minimum	25	4.6.2
Solids content, variation from preproduction test value, percent, maximum	10	4.6.2

TABLE I. *Physical requirements for coating compound*—(Cont'd.)

	Requirement	Test procedure
Tensile strength, psi, minimum	600	4.6.3
Ultimate elongation, percent, minimum	250	4.6.3
Adhesion, initial, stripping in three minutes by 10-pound load, inches, maximum	3	4.6.4.2
Adhesion after oven aging, stripping in three minutes by 8-pound load inches, maximum	3	4.6.4.3
Adhesion after water immersion, stripping in three minutes by 10-pound load, inches, maximum	3	4.6.4.4

3.4 Color. Unless otherwise specified in the contract or order (see 6.2), the coating material shall be of a gray color conforming to formula No. 20 or MIL–P–699 or type A of formula No. 5N of MIL–E–15129.

3.5 Marking. Each container of material shall be labeled giving adequate instructions for use and application of contents and safety precautions necessary. The labels shall be adhered to the containers with a water-resistant adhesive and the labels shall be waterproofed by a coating that does not smear or blur the markings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his

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own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction tests. Prior to production, preproduction tests shall be conducted on the coating material to determine conformance with all the requirements of this specification. All of the tests of 4.6 shall be conducted. The coating material shall be certified to be identical in composition to the material to be supplied later. The contractor shall not proceed with production under the contract or order without written approval of the preproduction test results by the Command or agency concerned. Preproduction tests are not required for material previously supplied under contract which has been tested and found to conform to this specification provided no changes have been made in production techniques or materials since such tests.

4.3 Sampling.

4.3.1 *Lot.* For purposes of sampling, examinations and tests, a lot shall consist of all the material produced in one plant under essentially the same conditions, not exceeding 2500 pounds, and offered for delivery at one time.

4.3.2 *Sampling for examination of containers.* A random sample of filled containers shall be taken in accordance with the sampling plan specified in table II for the examination specified in 4.4.

TABLE II. *Sampling for examination.*

Lot size Number of filled containers	Sample size Number of filled containers to be sampled	Number of nonconforming or defective containers			
		Major defects		Total defects	
		Accept	Reject	Accept	Reject
Up to 8	5	0	1	0	1
9 to 25	8	0	1	1	2
26 to 50	13	0	1	2	3
51 to 90	13	1	2	2	3
91 to 150	20	1	2	3	4
151 to 280	32	2	3	5	6
281 to 500	50	3	4	7	8
501 to 1200	80	5	6	10	11
1201 to 3200	125	7	8	14	15

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4.3.2.1 *Defects defined.* A major defect is a defect that is likely to result in failure, or to reduce materially the usability of the unit of product for its intended purpose. A minor defect is a defect that is not likely to reduce materially the usability of the unit of product for its intended purpose, or is a departure from established standards having little bearing on the effective use or operation of the unit. Total defects are major and minor defects combined.

4.3.3 *Sampling for quality conformance tests.* From each lot, two filled containers shall be taken at random for testing as specified in 4.5.

4.4 Examination of filled containers. Each filled container taken as specified in 4.3.2 shall be examined for defects including those specified in table III. Any defective container in the sample shall be cause for rejection and if the number of defective filled containers in the sample exceeds the applicable acceptance number of the sampling plan in table II, this shall be cause for rejection of the lot represented by the sample.

4.5 *Quality conformance tests.* The containers of material taken in accordance with 4.3.3 shall be two test samples. Each test sample shall be separately subjected to the tests specified in 4.6.1, 4.6.2. and 4.6.4.2.

4.5.1 *Rejection.* If any sample representing a lot is found to be not in conformance with this specification, it shall be cause for rejection and the lot represented by the sample shall not be considered acceptable.

4.6 Test procedures.[1]

4.6.1 *Viscosity.* Viscosity of the coating compound shall be determined with either a Brookfield or a McMichael viscosimeter, or equivalent. The test shall be run in duplicate using a new sample of material for each determination. Both the sample of coating compound and the apparatus shall be conditioned at the testing temperature of 27° ± 5°C. (80° ± 9°F.) for 4 hours before testing.

TABLE III. *Defects.*

Examine	Defects
Packaging	Container not type specified.
	Closure not type specified.
	Size not as specified.
	Inner seal of container not seated; damaged.
	Evidence of rust or corrosion.
	Bail or handle missing or damaged.
	Bail grip damaged.
	Container leaking.
Packing	Not as specified.
	Arrangement of container not as specified.
	Closing and strapping not as specified.
Marking	Interior or exterior markings (as applicable) omitted, illegible, incorrect, incomplete, or not as specified.
Fill	Container below the required or indicated fill.

4.6.2 *Solids content.*

4.6.2.1 *Procedure.* A covered low-form weighing bottle shall be weighed, and approximately ten grams of thoroughly mixed

[1] Unless specified otherwise herein all conditioning and testing of specimens shall be conducted at 27° ± 5°C. (80° ± 9°F.).

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coating compound shall be poured into the tared container, covered and weighed. After removing the cover, the container shall be placed in an oven at 100° ± 1.1°C. (212° ± 2°F.) until the sample reaches constant weight. The covered container with the sample shall be cooled in a desiccator before weighing. The test shall be run in duplicate.

4.6.2.2 *Calculation.* The percentage of total solids shall be calculated as follows:

$$\text{Total solids, percent} \quad \frac{\text{weight of residue}}{\text{weight of sample}} \times 100$$

4.6.3 *Tensile strength and ultimate elongation.*

4.6.3.1 *Specimen preparation.* Specimens shall be prepared by casting or applying multiple coats of the material approximately 3 inches wide by 8 inches long on silicone treated paper or other suitable release material. Film shall be allowed to dry at 27° ± 5°C. (80° ± 9°F.) for 24 ± ¼ hours, then stripped from the backing and placed in a circulating air oven at 70° ± 1.1°C. (158° ± 2°F.) for 24 ± ¼ hours. The final thickness of the strips shall be 0.020 to .035 inch thick. After removing the strips from the oven, condition the specimens at least 16 hours at 27° ± 5°C. (80° ± 9°F.) before testing.

4.6.3.2 *Testing.* Tensile strength and ultimate elongation of the strips shall be determined using methods 4111 and 4121 of FED–STD–601 and die III specimens.

4.6.4 *Adhesion.*

4.6.4.1 *Specimen preparation.* The adhesion specimens shall be prepared as follows: Clean the surfaces of nine steel panels, 3 by 6 by 1/16 inches either by grinding, wire brushing or preferably by sandblasting and then with solvent. Cut nine strips of cotton canvas, 2 by 6 inches. Apply two brush coats of the material over the steel panels (primed as necessary) and 5 to 6 inches of the canvas strips. Allow each coat to air dry ½ hour. Apply a third coat of the material to the steel and to the canvas and roll the two surfaces together while wet to insure good contact between the canvas and the coating applied on the panel. After drying for one hour, apply two additional coats over the assembly allowing ½ hour drying between coats. Allow all assembled specimens to air cure at 27° ± 5°C. (80° ± 9°F.) for 96 ± ½ hours before further treatment.

4.6.4.2 *Initial adhesion.* Initial adhesion tests shall be conducted on three of the specimens prepared as specified in 4.6.4.1. Just prior to testing, the specimens shall be cut through to the metal along 2 parallel lines, leaving a strip 1.00 ± 0.01 inch in width, running lengthwise through the center of the specimen. The portion outside of the one-inch strip shall be removed before testing. One end of the canvas shall be separated from the metal to permit the attachment of the weight specified in table I. The steel panel shall be supported at the ends in a horizontal position. The weight shall be attached to the free end of the canvas. Amount of stripping in inches in three minutes shall be noted. Results shall be averaged.

4.6.4.3 *Adhesion after oven aging.* Adhesion of three of the specimens shall be determined as specified in 4.6.4.2, after oven aging for 46 ± ¼ hours at 90° ± 1.1°C. (194° ± 2°F.). Allow specimens to cool to room temperature before testing adhesion.

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4.6.4.4 *Adhesion after immersion.* Three specimens shall be immersed in distilled water at 27° ± 1.1°C. (80° ± 9°F.) for 46 ± ¼ hours, remove and surface dried with paper towels, and then tested as specified in 4.6.4.2.

4.7 Examination of preparation for delivery. Random sample packs shall be examined to verify conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.* The coating compound or primer shall be furnished in 1 quart, 1 gallon or 5 gallon containers as specified (see 6.2).

5.1.1.1 *One-quart cans.* The 1 quart cans containing coating compound or primer shall conform to type V, class 2 of PPP–C–96. Exterior coating shall be required as specified in plan B.

5.1.1.2 *One-gallon cans.* The 1 gallon cans containing coating compound or primer shall conform to type V, class 2 of PPP–C–96. A wire handle or a formed bridge type handle securely affixed to the top shall be required. Exterior coating shall be required as specified in plan B.

5.1.1.3 *Five-gallon pails.* The 5 gallon pails containing coating compound or primer shall conform to type II, class 1 of PPP–P–704. Wire handles or bails shall be galvanized or protectively coated to resist corrosion.

5.1.2 *Level C.* Preservation and packaging shall be sufficient to afford adequate protection against deterioration and physical damage during shipment from the supply source to the using activity for early material use. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Coating compound or primer packaged in accordance with level A or C as specified (see 6.1) shall be packed in containers for overseas shipment in accordance with the appendix to PPP–C–96. Five gallon pails shall not require additional packing.

5.2.2 *Level B.* Coating compound or primer packaged in accordance with level A or C as specified (see 6.1) shall be packed in containers for domestic shipment in accordance with the appendix to PPP–C–96. Five-gallon pails shall not require additional packing.

5.2.3 *Level C.* Packing shall be accomplished in a manner which will insure acceptance by common carrier, at the lowest rate, and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early material use. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice when such meets these requirements.

5.3 Marking. In addition to any special marking required by the contract, or order, or herein (see 3.5 and 6.2), interior and exterior shipping containers shall be marked in accordance with MIL–STD–129 and shall include the month and year of manufacture and the lot number.

5.3.1 *Hazardous chemicals.* All packages containing hazardous chemicals shall have affixed thereto such warning labels and markings as may be required by the Interstate Commerce Commission Regulations, CFR Title 49, Parts 71-78 and the Manufacturing Chemists Association Manual L–1.

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6. NOTES

6.1 Surface preparation. In using this material to protect exposed steel surfaces from corrosion the steel surface to be coated should be prepared by grinding, wire brushing or preferably by sand or grit blasting. After abrading, the surface should be completely free from loose rust, scale, and oil. The surface should be solvent cleaned where oil or grease is present.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Color if other than specified (see 3.4).

(c) Size container required (see 5.1).

(d) Level of packaging and packing (see 5.1 and 5.2).

(e) Special marking, if required (see 5.3).

6.3 Preproduction.

6.3.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Custodians:

Army—MR

Navy—SH

Review activities:

Army—MR

Navy—SH

User activities:

Army—EL, GL, MI, MU

Navy—AS, MC, OS, YD

Preparing activity:

Navy—SH

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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